

[Dashb...](#) / [My co...](#) / [MA-224-...](#) / [I...](#) / [Test 3 \(topics 7-9: Recurrence relations, Introduction to group theory, The multiplicative group of inte...](#)

Status	Finished
Started	Thursday, 6 November 2025, 12:01 PM
Completed	Thursday, 6 November 2025, 12:30 PM
Duration	28 mins 55 secs
Marks	3.00/3.00
Grade	3.00 out of 3.00 (100%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade.**

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question 1

Correct

Mark 1.00 out of 1.00

The following sequence

$$p_n = -\frac{2 \cdot (-3)^{n+1}}{31}$$

is a particular solution to the recurrence relation

$$5 \cdot a_n - \frac{a_{n-1}}{2} = (-3)^n.$$

a) Find another solution to the recurrence relation, not equal to p_n :

$$a_n = \boxed{1/10^n - (2 \cdot (-3)^{n+1})/31}$$

Your last answer was interpreted as follows:

$$\frac{1}{10^n} - \frac{2 \cdot (-3)^{n+1}}{31}$$

The variables found in your answer were: $[n]$

Syntax guide: Your answer should be an expression which depends on the variable n , and *no other constants*. E.g. if you think the answer is

$$-7 \cdot \frac{2^n + 1}{3n},$$

you should type **$-7 \cdot (2^n + 1) / (3 \cdot n)$** .

b) Given the initial condition $a_0 = \frac{37}{31}$. Compute a_n for $n = 1$:

$$a_1 = \boxed{-149/310}$$

Your last answer was interpreted as follows:

$$\frac{-149}{310}$$

Question 2

Correct

Mark 1.00 out of 1.00

Let $(D_3, \cdot)$ be the dihedral group of symmetries of the regular triangle. We label the vertices of the triangle 1, 2, 3 in a counter-clockwise manner. The symmetry r_1 is rotation by 120 degrees, sending vertex 1 to vertex 2. The symmetry r_2 is rotation by 240 degrees. The element t_1 is the reflection symmetry that fixes vertex 1, and likewise for t_2 and t_3 .

The multiplication table for this group looks like

id	r1	r2	t1	t2	t3
r1	r2	id	t2	t3	t1
r2	.	r1	.	.	t2
t1	.	t2	.	.	r1
t2	.	t3	.	.	r2
t3	t2	t1	r2	r1	id

However, we have left out 9 entries in the places marked by dots.

Find the entire multiplication table for D_3 .

Syntax help: Remember that we compose symmetries from left to right. So if a and b are two symmetries, then the composite $a \cdot b$ is the symmetry that first applies a , and then b .

$r_2 \cdot t_1 =$	t3
$t_2 \cdot t_1 =$	r1
$t_1 \cdot t_1 =$	id
$r_2 \cdot t_2 =$	t1
$t_2 \cdot t_2 =$	id
$t_1 \cdot t_2 =$	r2
$r_2 \cdot r_1 =$	id
$t_2 \cdot r_1 =$	t1
$t_1 \cdot r_1 =$	t3

Your last answer was interpreted as follows:

t3

Your last answer was interpreted as follows:

r1

Your last answer was interpreted as follows:

id

Your last answer was interpreted as follows:

t1

Your last answer was interpreted as follows:

id

Your last answer was interpreted as follows:

r2

Your last answer was interpreted as follows:

id

Your last answer was interpreted as follows:

t1

Your last answer was interpreted as follows:

t3

Question 3

Correct

Mark 1.00 out of 1.00

Let $(P_9, \bullet_9)$ be the group of natural numbers coprime to 9 with group operation given by multiplication modulo 9. The multiplication table for this group looks like

1	2	4	5	7	8
2	4	8	1	5	7
4	8	7	2	1	5
5	.	2	7	.	.
7	.	1	8	.	.
8	.	5	4	.	.

However, we have left out 9 entries in the places marked by a dot "."

Compute the missing entries in the group multiplication table:

$5 \bullet_9 7 =$	8
$8 \bullet_9 7 =$	2
$7 \bullet_9 7 =$	4
$5 \bullet_9 8 =$	4
$8 \bullet_9 8 =$	1
$7 \bullet_9 8 =$	2
$5 \bullet_9 2 =$	1
$8 \bullet_9 2 =$	7
$7 \bullet_9 2 =$	5

Your last answer was interpreted as follows:

8

Your last answer was interpreted as follows:

2

Your last answer was interpreted as follows:

4

Your last answer was interpreted as follows:

4

Your last answer was interpreted as follows:

1

Your last answer was interpreted as follows:

2

Your last answer was interpreted as follows:

1

Your last answer was interpreted as follows:

7

Your last answer was interpreted as follows:

5

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 247

Signing code:

Your last answer was interpreted as follows:

54397

[◀ Test 2 \(topics 4-6: Languages, Machines, Regular expressions, Correct programs, Relations\)](#)

Jump to...